# LESSON - Concurrency in GUI Applications

## Qt Installation

- Download installer and register account
  https://www.qt.io/download-qt-installer
  https://login.qt.io/register
- Run installer (need to login)
  *Minimal installation (naming might be different depending on installer version)*

```
Qt
|
+-- Qt 6.9.x (or your downloaded version)
|    |
|    +-- Desktop
+- Build Tools
     |
     +-- CMake
     +-- Ninja
```

## CMake

- Installation
  Windows: https://cmake.org/download/

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https://doc.qt.io/qt-6/cmake-get-started.html

## Getting Started with QT

- Qt Getting Started
  https://doc.qt.io/qt-6/gettingstarted.html
- QWidget
  https://doc.qt.io/qt-6/examples-widgets.html
  |
  +-- https://doc.qt.io/qt-6/qtwidgets-widgets-scribble-example.html
  |
  +-- https://code.qt.io/cgit/qt/qtbase.git/tree/examples/widgets/widgets/scribble?h=6.9
- Building app
  Make sure CMakeLists.txt is in your project root folder

```
mkdir build
cd build
cmake .. -DCMAKE_PREFIX_PATH=<path to Qt CMake extension directory> (Ex: /Users/miko/Qt/6.9.2/macos/lib/cmake)
cmake --build .
```

- Executable file will be generated inside `build` folder

## GUI Application Components

- QApplication
  Runs the main event loop of the UI thread
  https://doc.qt.io/qt-6/qapplication.html#details
- QWidget
  Base class of other UI elements

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  This can also be used as a drawing canvas by
  - Overriding the paintEvent() function
  - Using its QPainter object for drawing
- Layout manager
  https://doc.qt.io/qt-6/layout.html
- Event Handling
  https://doc.qt.io/qt-6/eventsandfilters.html

## Simple Canvas w/ Threads Demo

This demo applies the multithreading concepts from previous lessons in a simple GUI app
simple_canvas.zip

**Zoom Lectures**

Setting up + Simple GUI demo (Passcode: +nRnY7%L)
GUI App Components (Passcode: Lvn.H@37)
Combining GUI and Multithreading (Passcode: =4zR^RX9)

This topic is closed for comments.